6-Month Data Structures & Algorithms Roadmap for FAANG Preparation

This roadmap is designed for students and professionals preparing for technical interviews at top tech companies such as Google, Amazon, Meta, Apple, Netflix, Microsoft, and other high-paying software engineering roles. The plan focuses on problem-solving, interview patterns, consistency, and mock interview practice.

Recommended Weekly Schedule

- Study DSA concepts: 2 hours/day
- Solve coding problems: 2–3 hours/day
- Revision: 1 hour/day
- Mock interview or contest: 1–2 times/week
- Target: 400–600 quality problems in 6 months
- Programming language: C++, Java, or Python

Month 1 — Foundations & Problem Solving

Topics to Study:

- Time and Space Complexity
- Arrays and Strings
- Recursion and Backtracking Basics
- Hashing
- Two Pointers
- Sliding Window
- Basic Sorting and Searching

Main Goals:

- Understand Big-O notation deeply
- Solve 100 easy-level problems
- Build coding speed and debugging habits

Practice Platforms	LeetCode Easy, HackerRank, GeeksforGeeks
Weekly Problem Target	15–20 problems/week

Month 2 — Core Data Structures

Topics to Study:

- Linked Lists
- Stacks and Queues
- Binary Search
- Bit Manipulation
- Prefix Sum
- Monotonic Stack

Main Goals:

- Master implementation and common patterns
- Start medium-level interview questions
- Improve optimal thinking

Practice Platforms	LeetCode Medium, NeetCode Roadmap
Weekly Problem Target	20–25 problems/week

Month 3 — Trees & Graphs

Topics to Study:

- Binary Trees
- BST
- Heaps/Priority Queue
- Trie
- Graph Traversal (BFS/DFS)
- Topological Sort
- Union Find (DSU)

Main Goals:

- Understand recursion deeply
- Solve graph traversal confidently
- Practice tree patterns repeatedly

Practice Platforms	LeetCode, CodeStudio, InterviewBit
Weekly Problem Target	25 problems/week

Month 4 — Dynamic Programming & Advanced Graphs

Topics to Study:

- 1D DP
- 2D DP
- Knapsack Patterns
- Longest Common Subsequence
- Shortest Path Algorithms
- Dijkstra and Floyd Warshall
- Greedy Algorithms

Main Goals:

- Develop DP intuition
- Learn state transitions
- Recognize recurring patterns

Practice Platforms	LeetCode Hard, AtCoder DP
Weekly Problem Target	25–30 problems/week

Month 5 — Advanced Interview Preparation

Topics to Study:

- Segment Trees
- Fenwick Trees
- Advanced Recursion
- System Design Basics
- SQL Basics
- Object-Oriented Design

Main Goals:

- Start mock interviews
- Improve communication skills
- Solve timed contests

Practice Platforms	Pramp, LeetCode Contests, Codeforces
Weekly Problem Target	30 problems/week

Month 6 — Revision & FAANG Interview Simulation

Topics to Study:

- Revision of all patterns
- Top FAANG Interview Questions
- Behavioral Questions
- Resume Review
- Mock Interviews
- Contest Practice

Main Goals:

- Simulate real interview environment
- Identify weak topics
- Refine communication and optimization

Practice Platforms	LeetCode Premium, Interviewing.io
Weekly Problem Target	Focused revision + mocks

Important Resources

- LeetCode Blind 75
- NeetCode 150
- Striver's A2Z DSA Sheet
- GeeksforGeeks DSA Self-Paced
- Codeforces for competitive programming
- Pramp and Interviewing.io for mock interviews

Final Advice:
Consistency matters more than intensity. Solve problems daily, revise patterns regularly, and focus on understanding rather than memorization. Maintain notes for important patterns and revisit failed questions every week.